St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Foundation to Medicine (FTM) | DLA 19 before FTM LEC 36

Introduction to the Nervous System Part 1 - General Overview

ask-anat@sgu.edu

Department of Anatomical Sciences

School of Medicine, St. George's University

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Objectives

SOM.MKII.BPM1.1.FTM.1.ANAT.1004	Describe the three basic functions of the nervous system.
SOM.MKII.BPM1.1.FTM.1.ANAT.1005	Describe the anatomical & functional divisions of the nervous system.
SOM.MKII.BPM1.1.FTM.1.ANAT.1006	Describe the components of the central nervous system (CNS).
SOM.MKII.BPM1.1.FTM.1.ANAT.1007	Describe the components of the peripheral nervous system (PNS).
SOM.MKII.BPM1.1.FTM.1.ANAT.1008	Describe the components and targets of the somatic nervous system.
SOM.MKII.BPM1.1.FTM.1.ANAT.1009	Describe the components of the autonomic nervous system.

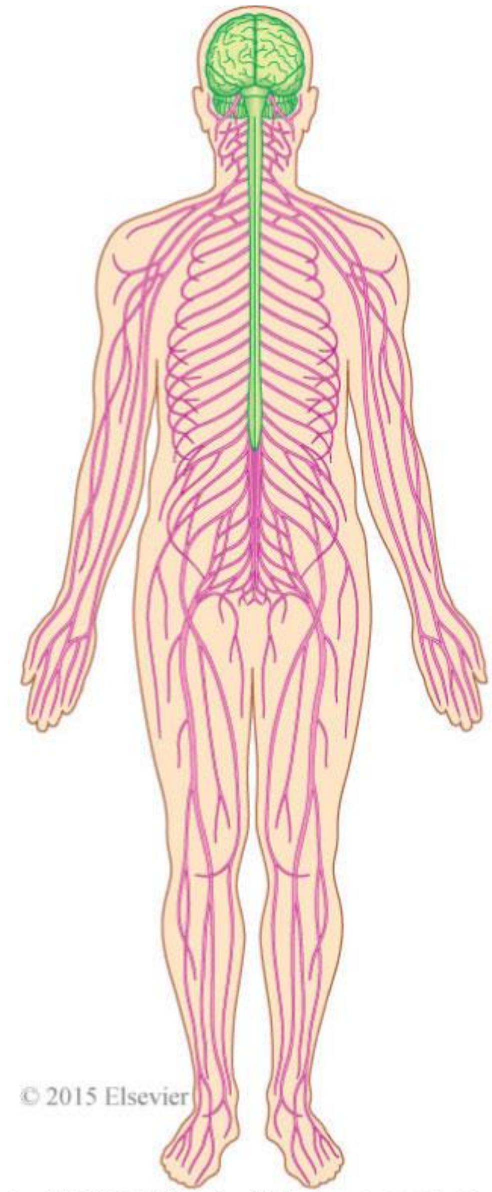
Nervous System Overview

Structural: **Central Nervous System (CNS)** — ***brain & spinal cord***

Peripheral Nervous System (PNS) — ***all nervous tissue outside the CNS***

Functional: **Somatic** — **Voluntary control:** **sensation** from body wall and limbs to CNS and **motor** neurons back to **skeletal muscle**

Visceral — **Involuntary control:** **sensation** from organs to CNS and **motor** neurons to **cardiac muscle, smooth muscle** and **glands**



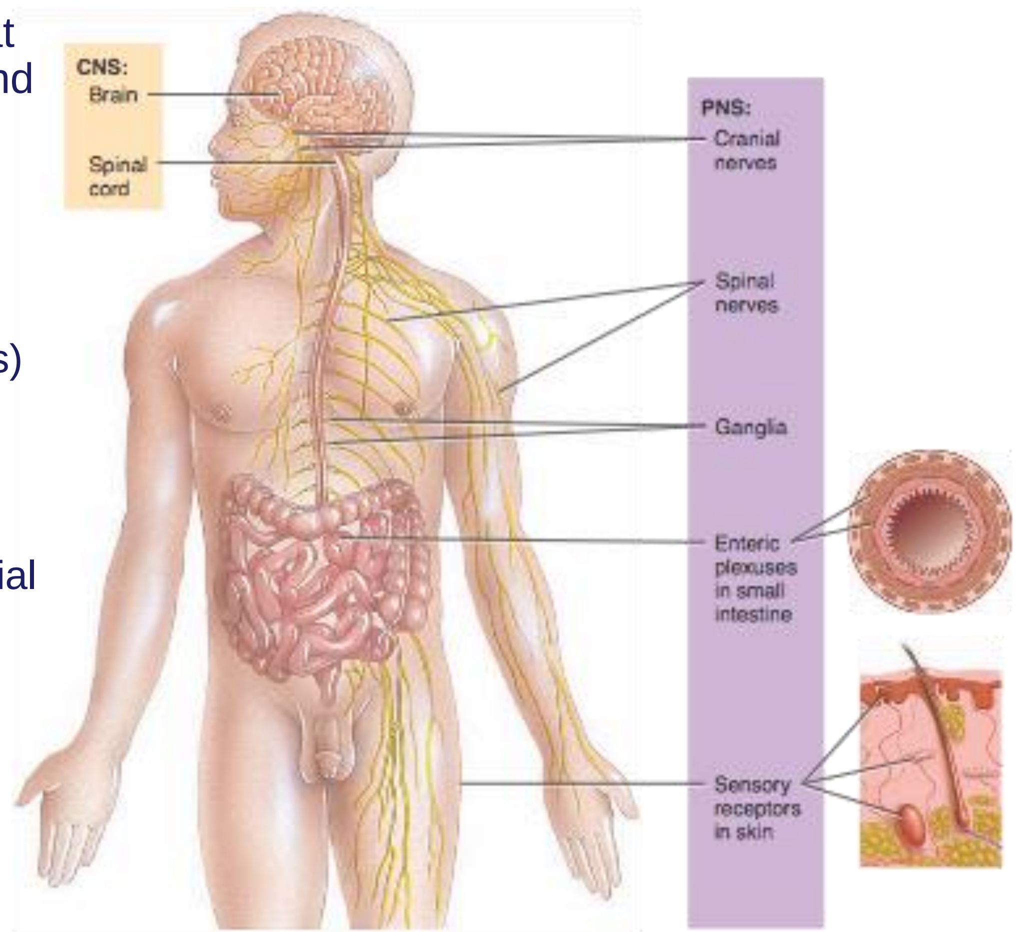
Components of the Nervous System

Central Nervous System

- Brain
- Spinal cord

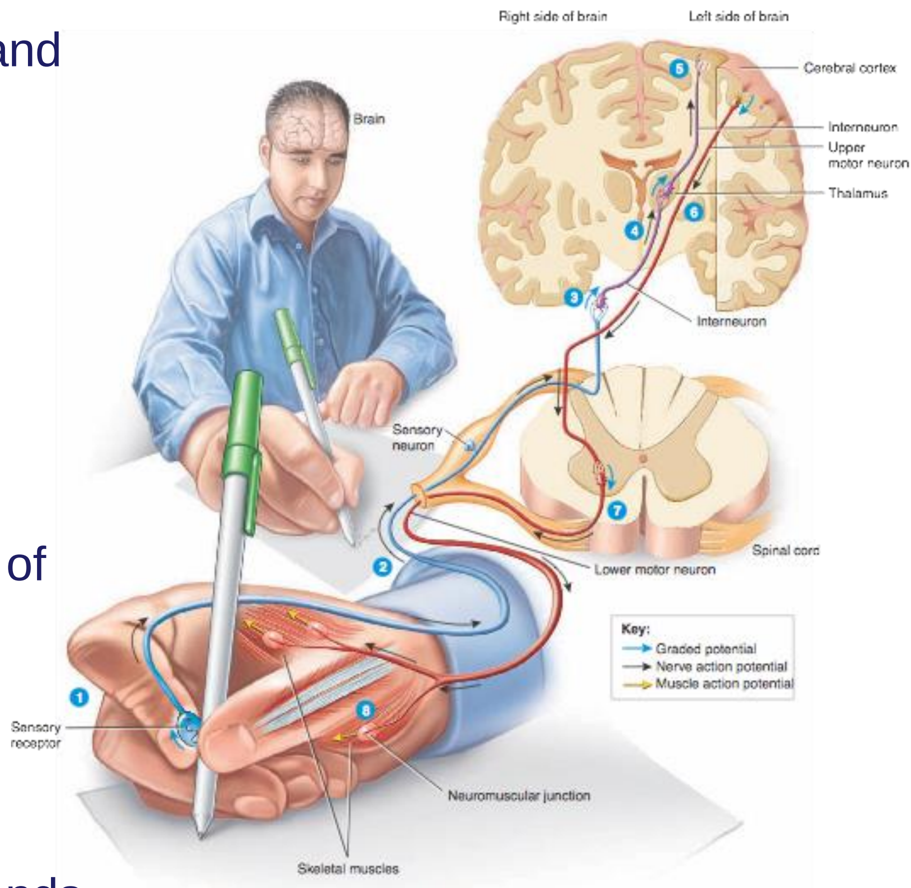
Peripheral Nervous System

- All nervous tissue outside of CNS that connects the CNS to the body wall and organs
- Cranial nerves
  - emerge from brain (12 pairs)
- Spinal nerves
  - emerge from spinal cord (31 pairs)
- Ganglia
  - consist of neuron cell bodies outside the brain and spinal cord
  - are associated with spinal & cranial nerves
- Enteric plexuses
  - help regulate digestive system
- Sensory receptors



Functions of the Nervous System

- **Sensory function**
  - detect internal stimuli
  - detect external stimuli
  - carries information into the brain and spinal cord
- **Integrative function**
  - processes sensory information
  - analyzes and stores information
  - makes decisions for appropriate responses
  - perception: conscious awareness of sensory stimuli
- **Motor function**
  - elicits an appropriate response
  - activates effector muscles and glands leading to contraction and secretion



Nervous System Overview

- **Peripheral Nervous System**
  - **Somatic**
    - **Motor (efferent)**
      - **Skeletal muscle**
        - **Intentional movement**
    - **Sensory (afferent)**
      - **Skin and fascia**
        - **P**ain
        - **P**ressure
        - **P**ropioception
        - **T**ouch
        - **T**emperature
  - **Visceral**
    - **Sensory (afferent)**
      - **P**ain:
      - Stretch/distension,
      - Inflammation,
      - Ischemia
      - Smooth muscle spasm Physiological Processes:
      - Temperature changes , Chemical changes, Hunger ,Nausea, Physiological reflexes
    - **Motor (efferent) - Autonomic**
      - **Glands**
      - **Smooth muscle**
      - **Cardiac muscle**
        - **Parasympathetic** (Rest and digest)
        - **Sympathetic** (Fight, flight or fright)
- **Central Nervous System**

SAME = Sensory Afferent – Motor Efferent

Nervous System Integration

Somatic

Receptors in **BODY WALL** for conscious sensation:
- pain
- pressure
- touch
- temperature
- proprioception

somatic sensory/afferent

somatic motor/efferent

Somatic

Voluntary control of:
- **skeletal muscle**

PNS

CNS

PNS

Visceral

Receptors in **ORGANS** and **TISSUES**:
- distension
- inflammation
- ischemia
- physiological changes
- abnormal muscle spasm

visceral sensory/afferent

visceral motor/efferent

(Autonomic Nervous System - ANS)

Visceral

Involuntary control of:
- **smooth muscle**
- **cardiac muscle**
- **glands**

Questions email:

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